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MIL-C-50785/13A (AR)
15 February 1993
SUPERSEDING
MIL-C-50785/13 (MU)
24 October 1975

MILITARY SPECIFICATION SHEET

CIRCUIT CARD ASSEMBLY: 12000013

(ATMOSPHERIC CORRECTION)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-50785.

REQUIREMENTS

1. Item Definition. This assembly is a single ended printed circuit board consisting of one capacitor, one set of resistors and an operational amplifier. The assembly is used to provide scaling factors within the Sight Current Generator Drawing Number 12000239 for air density characteristics.

2. Major component list. The components of this assembly shall be as specified on the Engineering Parts List (PL) 12000013.

3. Government furnished property list. Not applicable.

4. Characteristics.

4.1 Performance. The assembly shall be capable of performing the following requirements when assembled in accordance with Drawing Number 12000013.

a. Provide ballistic scaling for the Sight Current Generator Air Density Circuit.

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b. Provide a precision voltage source for the Sight Current Generator Air Density Circuit.

4.2 Input and output levels.

4.3 Input power. Input power shall be supplied to the assembly as specified in Table I.

TABLE I. Input voltages

Pin	Voltage
4	+15 Vdc ± 0.1 percent
14	-15 Vdc ± 0.1 percent
28	Ground

4.4 Output levels. The requirements specified in 4.5 shall be met when supplied with the specified voltage input.

4.5 Pin 17 output. The output between pin 17 and pin 28 shall be -3.938 Vdc ± 1.0 percent.

4.6 Resistance and capacitance values. The resistance and capacitance values shall be as specified in Table II. The maximum and minimum values shall be derated at temperature extremes of -40 degrees F and +160 degrees F per the applicable MIL specification covering each type resistor.

TABLE II. Resistance and capacitance values

Between Pin	Between Pin	Minimum Reading Ohm	Minimum Reading Microfarad	Maximum Reading Ohm	Maximum Reading Microfarad
15	TP2	2,090		2,310	
15	16	0.0000		0.0000	
13	18	0.0000		0.0000	
18	12	69,630		69,769	
12	11	87,112		87,287	
11	10	110,889		111,111	
10	9	147,852		148,148	
9	8	204,795		205,205	
8	7	283,716		284,284	
7	6	449,550		450,450	
6	5	774,225		775,775	
5	3	1,472,000		1,728,000	
3	2	4,692,000		5,508,000	
17	28		0.09		0.11
28	TP1	0.0000		0.0000	

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4.7 Physical characteristic. The physical characteristics shall be as specified in drawing 12000013 and the engineering parts list.

5. Test equipment. The test equipment listed below shall be used in the performance of the applicable methods of inspections as specified in 6.

5.1 Power supply (PS1). The power supply shall be +15 ± 0.1 Vdc, 100 milliamperes (ma or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

5.2 Power supply (PS2). The power supply shall be -15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

5.3 Digital ohmmeter. The digital ohmmeter shall have a resistance readout to five digits from 100 ohms to 10,000,000 ohms with 0.01 percent accuracy.

5.4 Digital voltmeter (DVM). The digital voltmeter shall have an accuracy and resolution of .01 percent or better from zero to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

5.5 Capacitance bridge. The capacitance bridge shall have a capacitance readout from 0.001 microfarad to 10 microfarad with one percent or better, accuracy and resolution.

6. Methods of inspection.

6.1 Test connections. Power supplies PS1 and PS2 shall be connected to the assembly under test in accordance with Figure 1 using receptacle 11726421.

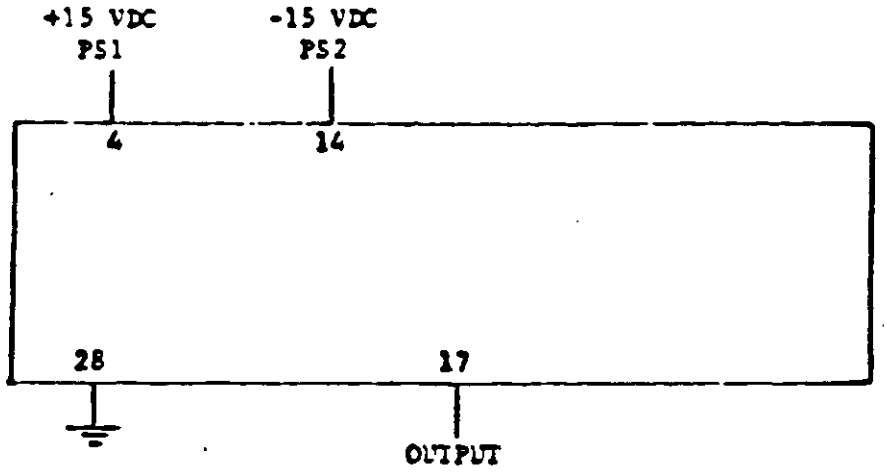


Figure 1. Test Connections

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6.2 Functional test procedure. Compliance with the requirements of 4.2 through 4.6 shall be verified as follows:

a. With inputs specified in Table I, the voltage measured with the DVM between pin 17 and pin 28 (ground) shall be -3.938 Vdc ± 1.0 percent.

b. With no voltages applied, the resistance measured with the digital ohmmeter between each set of pins shall be as specified in paragraph 4.6.

c. With no voltages applied, the capacitance measured with the capacitance bridge between each set of pins shall be as specified in paragraph 4.6.

7.0 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-50785/13A	15 FEBRUARY 1993

3. DOCUMENT TITLE: CIRCUIT CARD ASSEMBLY: 12000013 (ATMOSPHERIC CORRECTION)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

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